Fairness Aware Machine learning

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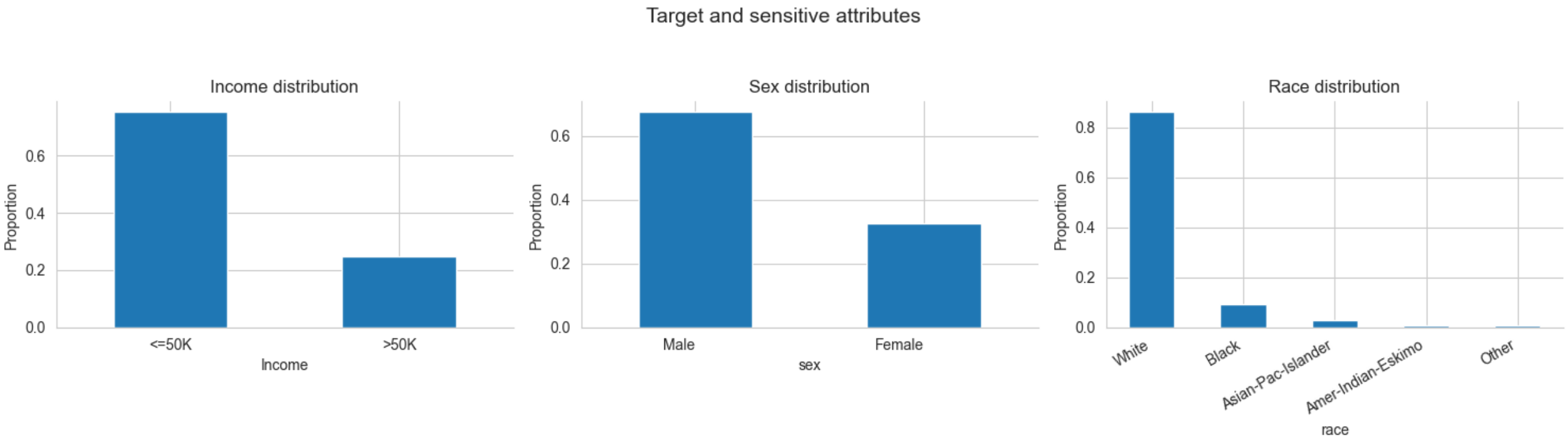
Dataset & Sensitive variable

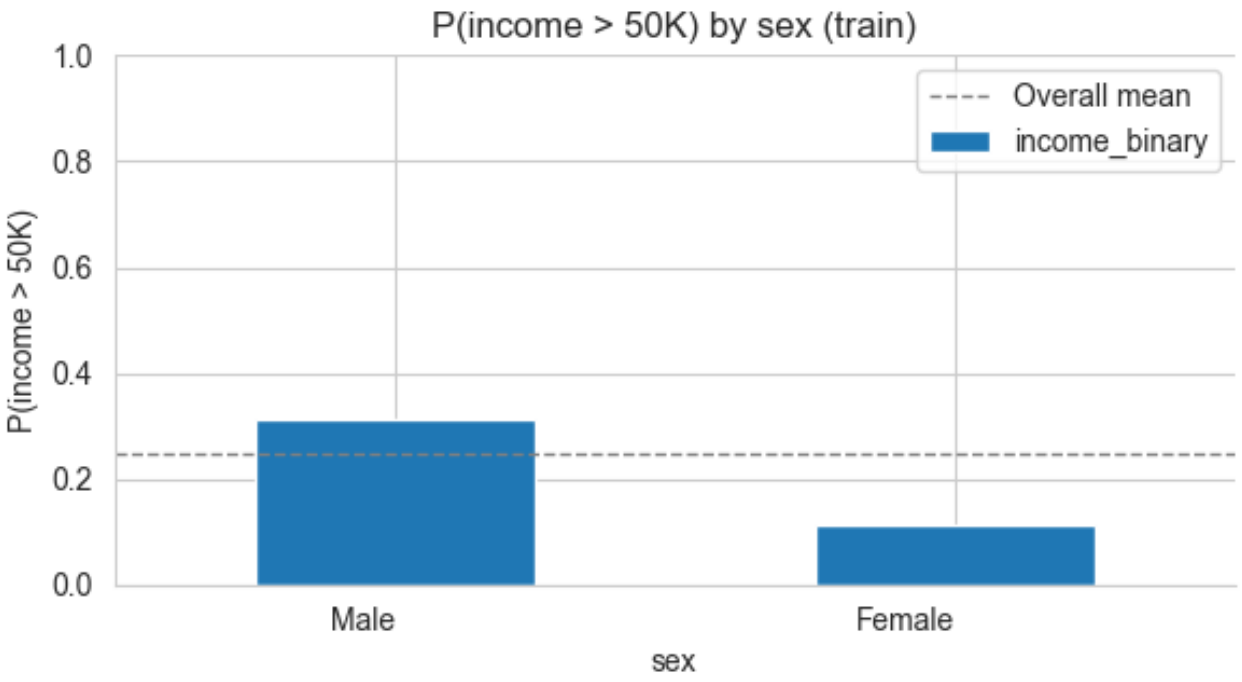
Dataset - Census income dataset

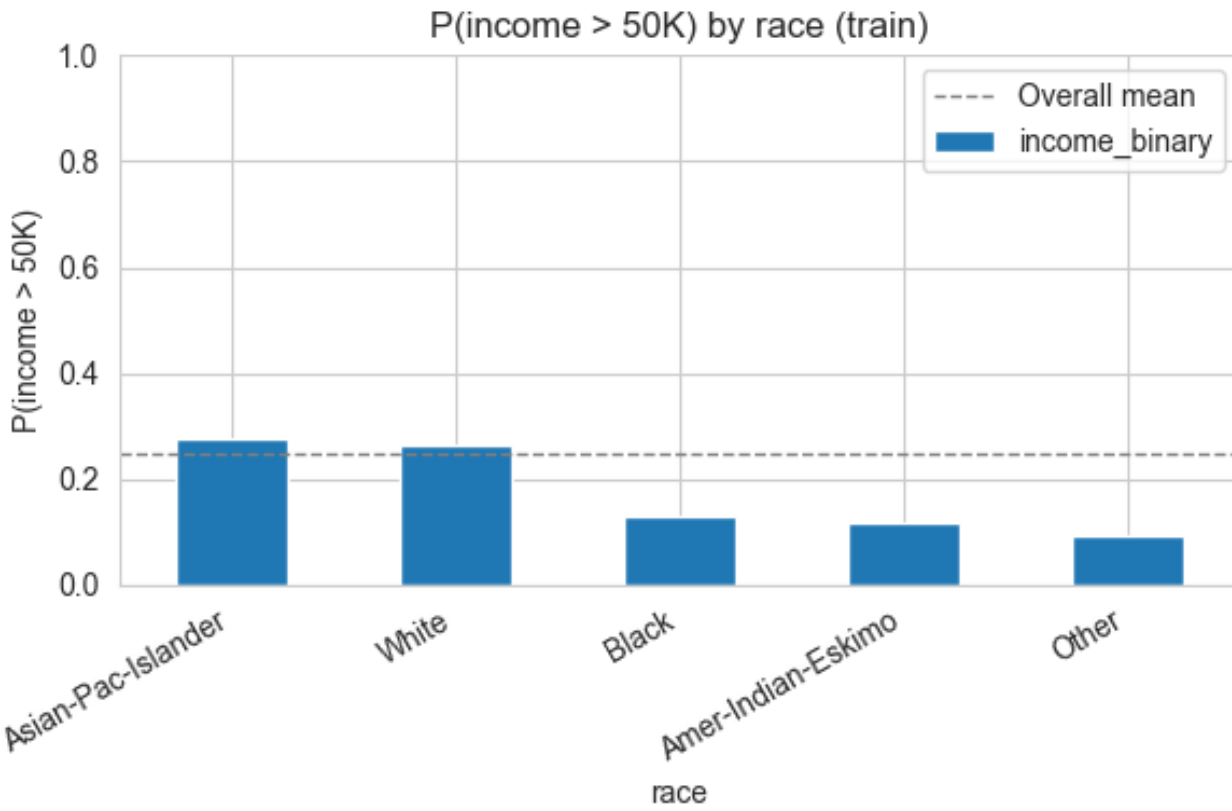
Sensitive variables considered – Race and Sex

EDA findings

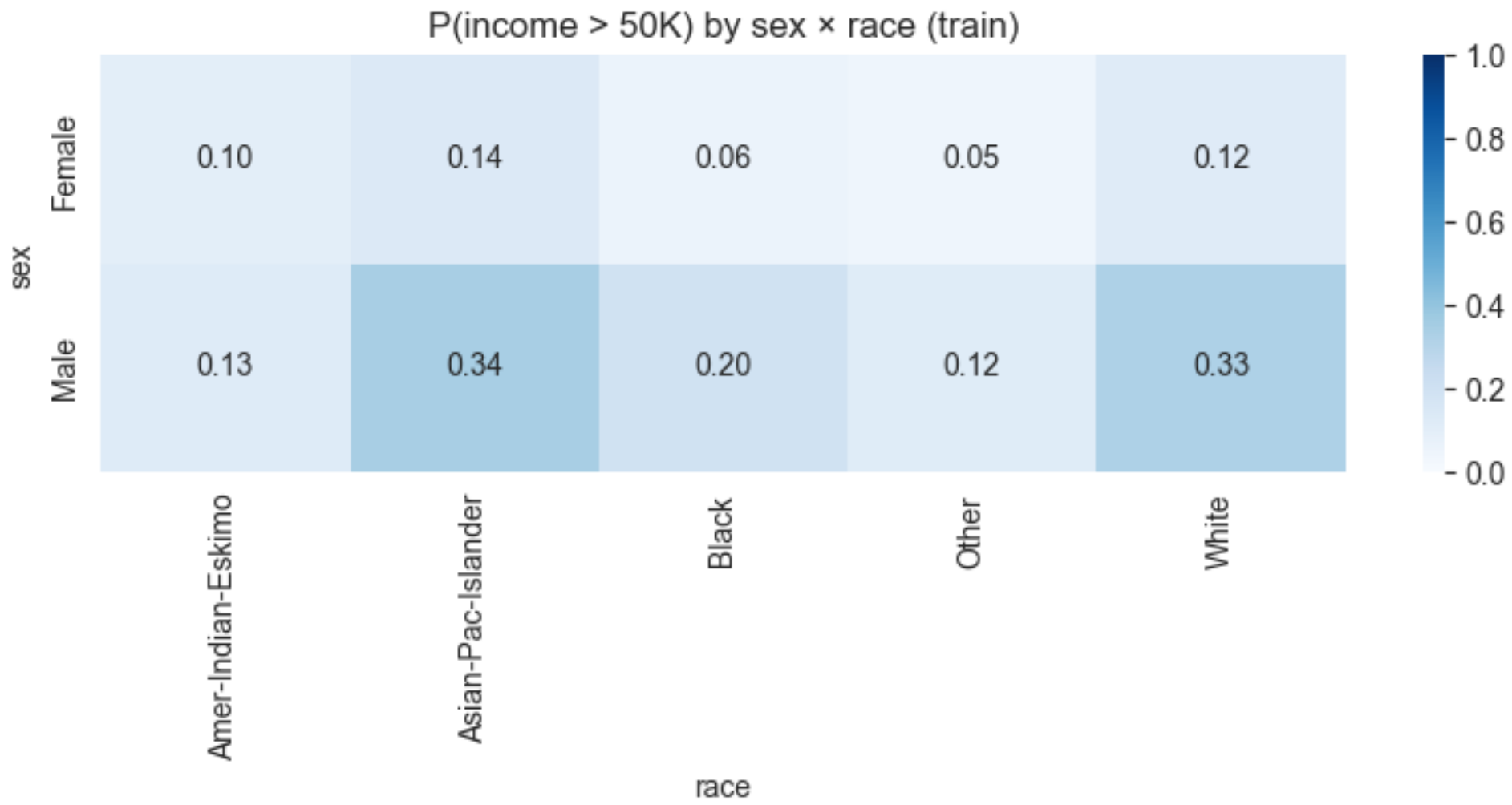


EDA findings





EDA findings



Fairness metrics

Demographic parity as fairness measurement method

"How often does each group receives income over the set threshold?"

Metrics:

Statistical parity difference (SPD)
"Gap in the positive outcome between privileged vs unprivileged groups"

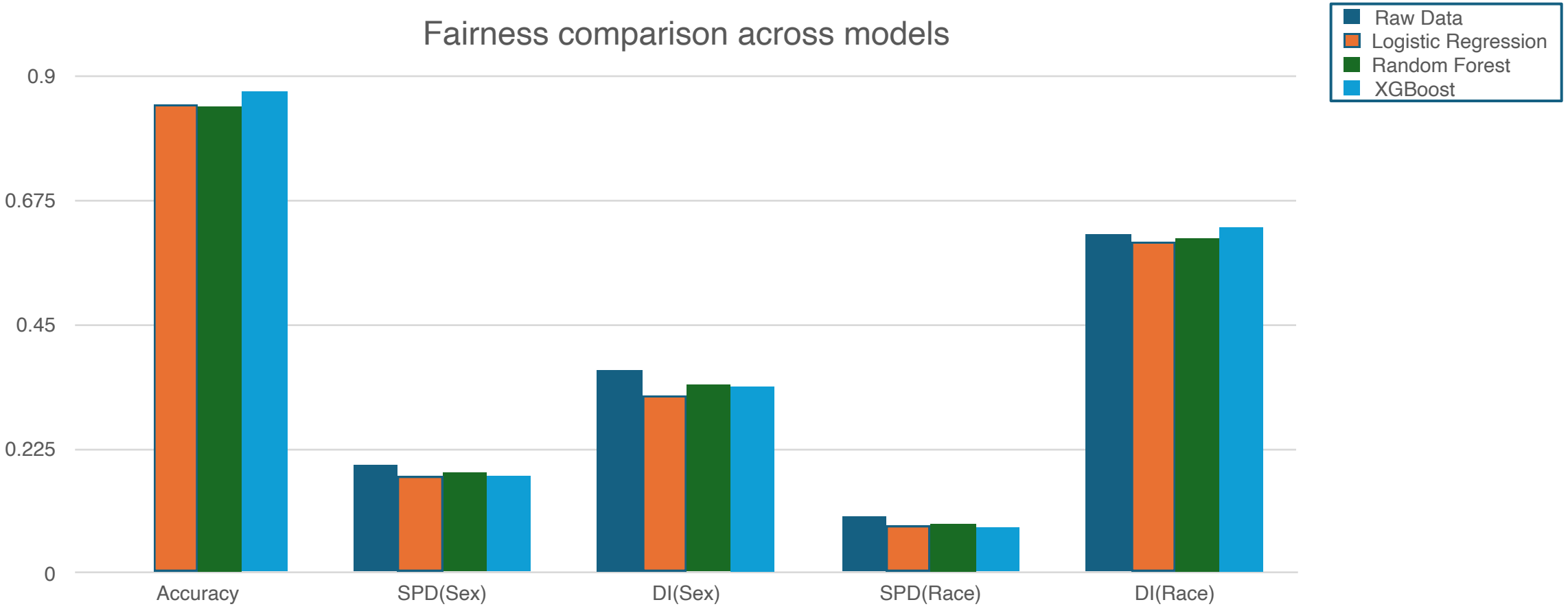
Disparate Impact (DI)
"Ratio – privileged vs unprivileged"

Raw Fairness

Sex: 'DP': 'Female': 0.11, 'Male': 0.30

'SPD': 0.196, 'DI': 0.365
Likelihood ratio 0.30/0.11 =2.73

Race: 'DP': ‘Non-White’: 0.158, 'White': 0.259

'SPD': 0.10, 'DI': 0.61
Likelihood ratio 0.26/0.16 =1.64

Model comparison – Accuracy & Fairness metrics